

# VaaniSetu Zero-Internet Neural Voice Mesh

System Architecture | Smart India Hackathon 2026 | Team AlomVilom

## 1. Overview

VaaniSetu is an offline-first Android system for multilingual voice communication in disaster and low-connectivity zones. Each phone converts speech to compact text on the device, protects it with error correction, and relays it phone-to-phone over a local wireless mesh. No cellular tower, internet or cloud API is needed at any stage. The full diagram is on the last page.

## 2. Architectural Layers

| Layer        | Components                                                                                                                                                                                          |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Application  | Push-to-talk, full-duplex phone mode, emergency distress broadcast, virtual channels, 10+ Indian languages, stealth mode & haptics, nearby device discovery, connection status, local message queue |
| Speech AI    | Silero VAD (2-stage), IndicConformerASR for on-device STT, VITS / IndicF5 for TTS, run through Sherpa-ONNX and ONNX Runtime Mobile                                                                  |
| Intelligence | Template Matcher (MiniLM / LaBSE) maps recognised speech to a 2-3 byte template ID; Brotli compresses unmatched text                                                                                |
| Reliability  | Reed-Solomon FEC encode/decode, TTL, duplicate suppression, retry and acknowledgement                                                                                                               |
| Network      | Mesh router (routing table, TTL decrement, smart forwarding, interface selection) over Wi-Fi Direct, Bluetooth SPP, optional LoRa/ESP32                                                             |
| Storage      | SQLite / Room: messages, contacts, channels, dictionary, settings                                                                                                                                   |
| Native       | C++ / JNI bridge for audio and inference-heavy components; Kotlin/Java for UI and services                                                                                                          |

## 3. End-to-End Data Flow

**Sender (Device A):** 16 kHz audio is captured on push-to-talk; a two-stage VAD (energy, then Silero) removes silence; IndicConformerASR transcribes on-device. The Template Matcher sends a 2-3 byte template ID on a match, or Brotli-compressed text otherwise. A header (sender, channel, language/urgency, TTL) is added, Reed-Solomon FEC is applied, and the mesh router picks an interface.

**Mesh:** Nodes keep a routing table, decrement TTL, suppress duplicates and forward smartly, so messages hop beyond direct radio range. Wi-Fi Direct covers about 50-200 m, Bluetooth SPP about 10-100 m, and optional LoRa/ESP32 about 1-5 km urban and 10-20 km line of sight.

**Receiver (Device B):** FEC decode, header parse, then template lookup or Brotli decompression, and storage in a Room/SQLite queue. Stealth Mode (proximity and ambient light sensors) chooses VITS speech playback or a Morse-like haptic alert.

## 4. Packet Format, SOS and Storage

- **Packet:** Sender ID, Channel ID, Lang/Urgency, TTL, Type, Payload (template ID or Brotli text).
- **SOS:** a high-priority distress broadcast to all nearby devices, bypassing silent mode and Do Not Disturb.
- **Local storage:** messages, contacts, channels, template dictionary and settings in SQLite.

## 5. Repository and Roadmap

- **Repo:** VaaniSetu/ (Android source), Apk/, docs/, submission/, screenshots/, README.md.
- **Phase 2:** LoRa/ESP32 hardware, advanced template compression, more languages and dialects.

## VaaniSetu Architecture Diagram

# VaaniSetu – Zero-Internet Neural Voice Mesh

Offline multilingual voice communication for disaster and low-connectivity zones



AlomVilom

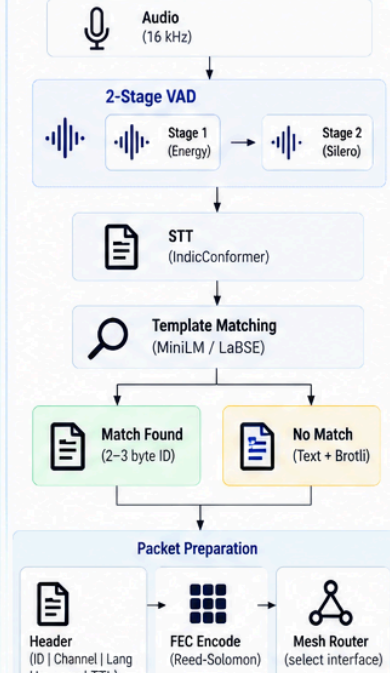
### 2 Technology Stack (On-Device)



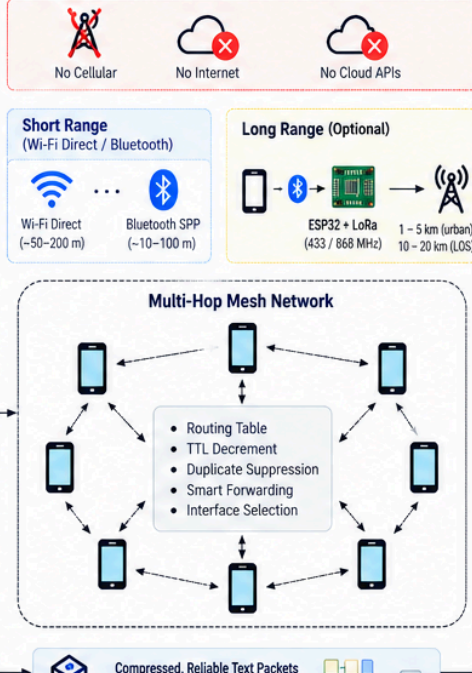
### 3 Application Features

- Full-Duplex Phone Mode
- Push-to-Talk (PTT)
- Emergency Distress Broadcast
- Virtual Channels
- 10+ Indian Languages
- Stealth Mode & Haptics
- Nearby Device Discovery
- Connection Status
- Local Message Queue

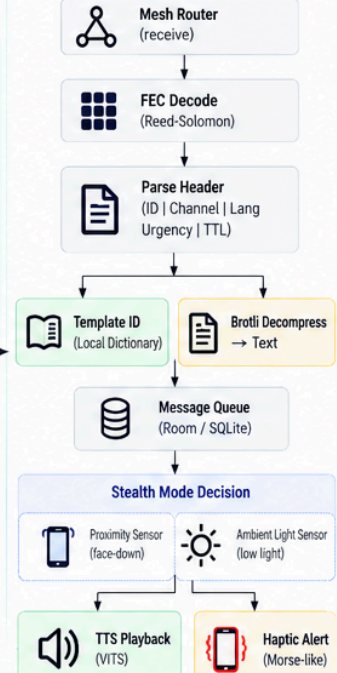
### Device A – Sender (Android Phone)



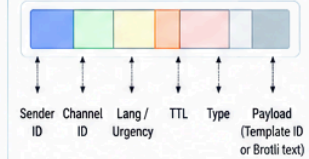
### 4 Communication Layer (Local Wireless Mesh)



### Device B – Receiver (Android Phone)



### 6 Data Format



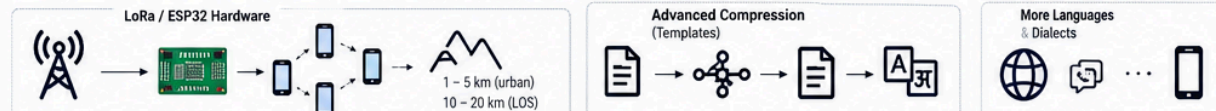
### 7 Local Storage (On Device)

- Messages
- Contacts
- Channels
- Dictionary
- Settings

### 8 Routing & Networking



### 9 Phase 2 / Planned Extensions



### 10 Repository Structure

